ADVANCE PART TEST-01

(JEE Advanced Pattern)

Subject		Syllabus
Mathematics	:	FOM 1 + Quadratic equations
Physics	:	Mathematical tools
Chemistry	:	Atomic structure
TOTAL MARKS ::180		

PART-I : MATHEMATICS

SECTION – 1 : (Maximum Marks : 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct
- For each question, marks will be awarded in one of the following categories :
  - Full Marks : **+4** If only (all) the correct option(s) is (are) chosen.
  - Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.
  - Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct.
  - Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.
  - Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).
  - Negative Marks : **–2** In all other cases.
- For example, in a question, if (A),(B) and (D) are the ONLY three options corresponding to correct answers, then :
  - Choosing ONLY (A),(B) and (D) will get +4 marks
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  - Choosing ONLY (B) will get +1 marks
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  - Choosing any other combination of options will get –2 mark

1. Identify which of the following statement(s) are '**CORRECT**' ?

(A) For $ax^2 + bx + c = 0$, $(a \neq 0)$ if $4a + 2b + c = 0$, then roots are 2 and $\frac{c}{2a}$.

(B) If α is repeated root of $ax^2 + bx + c = 0$, $(a \neq 0)$ then $ax^2 + bx + c = (x - \alpha)^2$.

(C) For $ax^2 + bx + c = 0$, $(a \neq 0, a,b,c \in Q)$ imaginary roots occur in conjugate pair only.

(D) If $f(x) = ax^2 + bx + c$, $(a \neq 0)$ has finite maximum value and both roots of $f(x) = 0$ are of opposite sign, then $f(0) > 0$.

Space for Rough Work)

2. If 'a' and 'b' are distinct positive integers and the quadratic equations $(a-1)x^2 - (a^2+2)x + (a^2+2a) = 0$ and $(b-1)x^2 - (b^2+2)x + (b^2+2b) = 0$ have a common root. Then which of the following can be TRUE?

(A) $a^2 + b^2 = 45$ (B) $a = 2b$ (C) $b = 2a$ (D) $ab = 18$

3. The equation $\log_{x^2} 16 + \log_{2x} 64 = 3$ has :

(A) one irrational solution (B) no solution is a prime number

(C) two real solutions (D) one integral solution

4. Consider the quadratic expression , $f(x) = x^2 - 8x + 12 + 4K - K^2$, $K \in R$, then the correct options are:

(A) If both roots of $f(x) = 0$ are positive then $K \in (-2, 6)$.

(B) If both roots of $f(x) = 0$ are negative then $K \in (6, \infty)$.

(C) Maximum value of $f(4)$ is 0.

(D) Minimum value of $f(4)$ is 0.

5. The solution of $(x+y)^{2/3} + 2(x-y)^{2/3} = 3(x^2-y^2)^{1/3}$ and $3x - 2y = 13$ are (x_1, y_1) and (x_2, y_2) $(x_1 > x_2)$ then

(A) $x_1 + y_1 = 16$ (B) $3x_2 + 3y_2 = 13$ (C) $x_1 y_1 = 63$ (D) $x_2 y_2 = \frac{19}{3}$

6. Which of the following statement(s) is/are true?

(A) $\log_{10} 2$ lies between $\frac{1}{4}$ and $\frac{1}{3}$

(B) $\log_{\operatorname{cosec}\left(\frac{\pi}{6}\right)}\left(\cos\frac{\pi}{3}\right) = -1$

(C) $e^{\ln(\ln 3)}$ is smaller than 1

(D) $\log_{10} 1 + \frac{1}{2}\log_{10} 3 + \log_{10}\left(2+\sqrt{3}\right) = \log_{10}\left(1+\sqrt{3}+\left(2+\sqrt{3}\right)\right)$

Space for Rough Work)

SECTION – 2 : (Maximum Marks : 24)

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7. Suppose a and b are rational numbers and α, β be roots of $x^2 + 2ax + b = 0$ then equation with rational coefficients one of whose roots is $\alpha + \beta + \sqrt{\alpha^2 + \beta^2}$, is

(A) $x^2 + 2ax + 2b = 0$ (B) $x^2 + 4ax - 2b = 0$ (C) $x^2 - 4ax + 2b = 0$
(D) $x^2 - 4ax - 2b = 0$ (E) $x^2 + 4ax + 2b = 0$

8. The number of integral solutions of $\dfrac{x^2(3x-4)^3(x-2)^4}{\ln(2x+1)(x-5)^5(2x-7)^6} \le 0$ is:

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

9. If x_1, x_2, x_3, x_4, x_5 are roots of $x^5 - 3x^3 - 1 = 0$ then $\dfrac{\sum\left(x_i^{100} - x_i^{95}\right)}{\sum x_i^{98}} =$

(A) 10 (B) 2 (C) 15 (D) 6 (E) 3

10. The least integral value of 'a' such that $(a - 2)x^2 + 8x + a + 4 > 0, \forall x \in R$ is

(A) 3 (B) 5 (C) 4 (D) 6 (E) 7

11. If value of k for which $(k^2 - 3k + 2)x^2 + (k^2 - 5k + 6)x + (k^2 - 4) = 0$ is an identity in x is λ then value of $\lambda^4 - 1$ is

(A) 12 (B) 13 (C) 15 (D) 11 (E) 14

Space for Rough Work)

12. Number of value(s) of `x' satisfying the equation : $(x)^{\log_{\sqrt{x}}(x-3)} = 9$ is/are

(A) 0 (B) 1 (C) 2 (D) 6 (E) 3

13. Number of positive integers 'n' such that $\frac{(n+1)^2}{n+7}$ is an integer.

(A) 2 (B) 4 (C) 6 (D) 8 (E) 10

14. The solution set of the inequality $\log_{(2x-3)}(3x-4) > 0$ is equal to $(\alpha,\beta) \cup (\gamma, \infty)$, then $\alpha\beta\gamma$ is equal to

(A) 2 (B) 3 (C) 5 (D) 6 (E) 0

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Paragraph for Question Nos. 15 to 16

Consider the inequality

$$\frac{(1-x)^3(x+2)^2(x^2-2x-3)}{(x-1)(x^2+1)(x)^3(x+5)} \geq 0$$

Let number of negative integer solutions is α and number of positive integer solutions is β.

15. The value of $\log_{\beta}\alpha$ is –

(A) $\log_3 4$ (B) $\log_4 3$ (C) 2 (D) $\frac{1}{2}$

16. Number of real solution of $x^2 - \alpha x + \beta\sqrt{2} = 0$ is

(A) 0 (B) 1 (C) 2 (D) 3

Paragraph for Question Nos. 17 to 18

Let x_1, x_2, x_3, x_4 be the roots (real or complex) of the equation $x^4 + ax^3 + bx^2 + cx + d = 0$.
If $x_1 + x_2 = x_3 + x_4$ and $a, b, c, d \in R$, then

17. If a = 2, then the value of b – c is

(A) –1 (B) 1 (C) –2 (D) 2

18. Let b + c = 1 and $a \neq -2$, if a is real then the possible values of 'c' are given by the range

(A) $\left(-\infty, \frac{1}{4}\right)$ (B) $(-\infty, 3)$ (C) $(-\infty, 1)$ (D) $(-\infty, 4)$

Space for Rough Work)

PART-II (भाग-II) : PHYSICS (भौतिक विज्ञान)

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  - Choosing ONLY (B) and (D) will get +2 marks
  - Choosing ONLY (A) will get +1 marks
  - Choosing ONLY (B) will get +1 marks
  - Choosing ONLY (D) will get +1 marks
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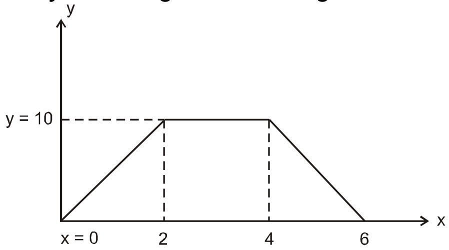
19. Which of the following are vector quantities:
(A) Force (B) Electrical current (C) Small area (D) Work

20. An aeroplane moves 5 km towards east, 4 km towards south and 3 km vertically upward direction. The unit vector along east, north and vertically upward direction are respectively $\hat{i}$, $\hat{j}$ and $\hat{k}$. If the displacement vector of the aeroplane is $\vec{S}$, then choose the correct alternatives :
(A) $\vec{S} = 5\hat{i} - 4\hat{j} + 3\hat{k}$
(B) $\vec{A} = 10\hat{i} - 8\hat{j} + 6\hat{k}$ will be parallel to $\vec{S}$
(C) $\vec{B} = 15\hat{i} + 12\hat{j} + 9\hat{k}$ will be parallel to $\vec{S}$
(D) $\vec{C} = 2\hat{i} + 4\hat{j} + 2\hat{k}$ will be perpendicular to $\vec{S}$

Space for Rough Work

21. Vector sum of the forces of 5N and 4N can be :
(A) 10 N (B) 1 N (C) 3 N (D) 9 N

22. The graph between y v/s x is given in the figure . Choose the correct alternative(s):



(A) $\left(\frac{dy}{dx}\right)_{at\ x=3} = 4$ (B) $\left(\frac{dy}{dx}\right)_{at\ x=5} = -5$ (C) $\int_{x=0}^{x=6} y\,dx = 40$ (D) $\int_{x=2}^{x=6} y\,dx = 30$

23. Which of the following sets of displacements might be capable of bringing a car to its returning point?
(A) 5, 10, 30 and 50 km (B) 5, 9, 9 and 16 km
(C) 40, 40, 150 and 90 km (D) 10, 20, 40 and 70 km

24. The value(s) of x which give maxima or minima of the function $f(x) = 7x^5 - 35x + 12$ is/are:
(A) 0 (B) 1 (C) – 1 (D) $\frac{1}{2}$

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25. If speed of a particle is 30 m/sec and its direction is from A (5,0,3) to point B (3,2,2) then its velocity vector will be :
(A) $20\hat{i} + 20\hat{j} + 10\hat{k}$ (B) $20\hat{i} - 20\hat{j} + 10\hat{k}$ (C) $-20\hat{i} + 20\hat{j} + 10\hat{k}$
(D) $-20\hat{i} + 20\hat{j} - 10\hat{k}$ (E) $2\hat{i} + 2\hat{j} + 1\hat{k}$

Space for Rough Work

26. The radius 'r' of a cylinder increases with time at a constant rate of 2m/sec and its height decreases with time at constant rate of 3 m/sec. The rate of change of volume of cylinder at a time when radius and height of cylinder are 3m and 2m respectively will be :

(A) 3π m^3/sec. (B) -3π m^3/sec. (C) 5π m^3/sec.
(D) -10π m^3/sec. (E) 0

27. $\int_{x=0}^{x=4} \frac{dx}{20-3x}$ will be :

(A) $\frac{1}{3}\ell n\left(\frac{5}{2}\right)$ (B) $\frac{1}{3}\ell n\left(\frac{2}{5}\right)$ (C) $3\ell n\left(\frac{5}{2}\right)$
(D) $3\ell n\left(\frac{5}{4}\right)$ (E) $\frac{1}{3}\ell n\ 5$

28. If y = sin(x) + $\ell n(x^2)$ + e^{2x} then $\frac{dy}{dx}$ will be :

(A) $\cos x + \frac{2}{x} + e^{2x}$ (B) $\cos x + \frac{2}{x} + 2e^{2x}$ (C) $-\cos(x) + \frac{2}{x^2} + e^{2x}$
(D) $-\cos x - \frac{2}{x^2} + e^{2x}$ (E) 0

29. Determine that vector which when added to the resultant of $\vec{P} = 2\hat{i} + 7\hat{j} - 10\hat{k}$ and $\vec{Q} = \hat{i} + 2\hat{j} + 3\hat{k}$ gives a unit vector along X-axis.

(A) $-2\hat{i} - 9\hat{j} + 7\hat{k}$ (B) $+2\hat{i} + 9\hat{j} - 7\hat{k}$ (C) $-2\hat{i} + 7\hat{j} + 9\hat{k}$
(D) $+2\hat{i} - 5\hat{j} + 3\hat{k}$ (E) $-4\hat{i} - 9\hat{j} + 7\hat{k}$

30. The position of a particle moving along the y - axis is given as $y = 3t^2 - t^3$ where y is in metres and t is in sec. The time when the particle attains maximum positive position will be

(A) 1.5 sec. (B) 4 sec. (C) 2 sec.
(D) 3 sec. (E) 1 sec.

Space for Rough Work

31. Find value of $\sin^2 15^\circ + \sin^2 645^\circ$:

(A) $\frac{1}{2}$ (B) 1 (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{3}$ (E) $\frac{1}{4}$

32. If $\sin\theta = 1/3$, then $\cos\theta$ will be -

(A) $\frac{8}{9}$ (B) $\frac{4}{3}$ (C) $\frac{2\sqrt{2}}{3}$ (D) $\frac{3}{4}$ (E) $\frac{1}{2}$

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Paragraph for Questions 33 and 34

There are two vectors $\vec{A}$ & $\vec{B}$. The x and y components of vector $\vec{A}$ are 4 m and 6 m respectively. The x, y components of vector $\vec{A}+\vec{B}$ are 10 m and 9 m respectively. Find :

33. The magnitude of $\vec{B}^2$.

(A) 45 (B) 60 (C) 75 (D) 50

34. Angle between $\vec{B}$ and X-axis.

(A) $\tan^{-1}\frac{1}{3}$ (B) $\tan^{-1}\frac{1}{2}$ (C) $\tan^{-1}\frac{1}{4}$ (D) $\tan^{-1}\frac{1}{\sqrt{2}}$

Paragraph for Questions 35 and 36

If a function is written as $y_1 = \sin(4x^2)$ & another function is $y_2 = \ln(x^3)$ then :

35. dy_1/dx will be :

(A) $8x\cos(4x^2)$ (B) $\cos(4x^2)$ (C) $-\cos(4x^2)$ (D) $-8x\cos(4x^2)$

36. dy_2/dx will be :

(A) $1/x^3$ (B) $3/x$ (C) $-1/x^3$ (D) $3/x^2$

Space for Rough Work

PART-III : CHEMISTRY

Atomic masses : [H = 1, D = 2, Li = 7, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, Si = 28, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Cu = 63.5, Zn = 65, As = 75, Br = 80, Ag = 108, I = 127, Ba = 137, Hg = 200, Pb = 207]

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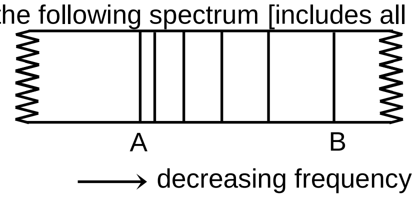
37. The energy level of a hypothetical H-like species is given by = $\frac{-20z^2}{(n+1)^2}$ ev

A sample of such atoms has electron in a particular upper energy level that jumps to ground state giving the following spectrum [includes all possible transitions]

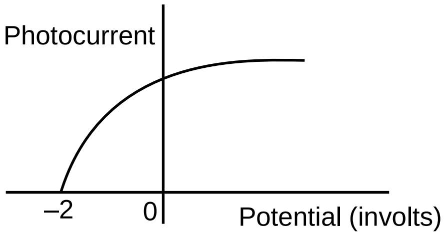


If wavelength A has λ = 73.8 nm, then the correct information (s) is/are :

(A) z must be 2
(B) z must be 3
(C) Wavelength of 'B' line is 405.2 nm
(D) Wavelength of 'B' line is 688.89 nm

Space for Rough Work

38. If light of wavelength 3100Å is used in the experiments, then following graph is observed (hc = 1240 nm –ev)



(A) Threshold wavelength is 620 nm
(B) An electron may be emitted with KE = 1ev in the above experiment.
(C) An electron may be emitted with KE = 4eV in the above experiment.
(D) Threshold wavelength is 310 nm

39. An α particle and a proton are projected towards the same nucleus in different experiments. The ratio of their distance of closest approach is :-
(A) 2 : 1 if their initial kinetic energy was same
(B) 8 : 1 if their initial momentum was same
(C) 1 : 2 if their initial speed was same.
(D) 1 : 1 if their initial kinetic energy was same.

40. Using following information, which of the following relations is/are correct for n^{th} orbit of a single electron species having atomic number Z.
$r_{n,z}$ = Radius of n^{th} orbit.
$V_{n,z}$ = Velocity of electron in n^{th} orbit.
$E_{n,z}$ = Magnitude of total energy of electron.
$K_{n,z}$ = Magnitude of kinetic energy of electron.
$P_{n,z}$ = Magnitude of potential energy of electron.
$F_{n,z}$ = Frequency of electron.
$T_{n,z}$ = Time period of electron.

(A) $\frac{F_{1,2}}{F_{1,1}} = \frac{r_{2,3}}{r_{1,3}}$ (B) $\frac{|E_{3,2}|}{K_{3,1}} = \frac{T_{1,1}}{T_{1,2}}$ (C) $\left(\frac{1}{\sqrt{2}}\right)\left(\sqrt{\frac{P_{2,1}}{E_{3,1}}}\right) = \frac{V_{2,3}}{V_{1,1}}$ (D) $\frac{E_{1,4}}{E_{2,2}} = 4\left(\frac{V_{2,5}-V_{2,1}}{V_{2,3}-V_{2,2}}\right)$

Space for Rough Work

41. Which of the following wave functions may not be appropriate for 3d orbital of a H-like species?

(A) $\Psi = \left(\frac{3}{4\pi}\right)^{\frac{1}{2}} \sin\theta \sin\phi \times \frac{1}{9\sqrt{6}}\left(\frac{1}{a_0}\right)^{3/2}\left(4 - \frac{2r}{3a_0}\right)\left(\frac{2r}{3a_0}\right)e^{-\frac{r}{3a_0}}$

(B) $\Psi = \left(\frac{1}{4\pi}\right)^{\frac{1}{2}} 2 \times \left(\frac{1}{a_0}\right)^{3/2} e^{-\frac{r}{a_0}}$

(C) $\Psi = \left(\frac{15}{4\pi}\right)^{\frac{7}{2}} \sin\theta \cos\phi \frac{1}{9\sqrt{30}}\left(\frac{1}{a_0}\right)^{3/2}\left(\frac{4r^2}{9a_0^2}\right)e^{-\frac{r}{3a_0}}$

(D) $\Psi = \left(\frac{3}{4\pi}\right)^{\frac{1}{2}} \sin\theta \cos\phi \frac{1}{2\sqrt{2}}\left(\frac{1}{a_0}\right)^{3/2}\left(1 - \frac{2r}{3a_0}\right)\left(\frac{9r^2}{4a_0}\right)e^{-\frac{r}{a_0}}$

42. Which of the following statements correspond(s) to Pauli's exclusion principle ?

(A) No 2 electrons in an atom can have the same value for all 4 quantum numbers.

(B) The most stable arrangement of electrons in subshells is the one with the greatest no. of parallel spins.

(C) An orbital can hold a maximum of 2 electrons and the electrons must have opposite spins.

(D) Pairing of electrons in degenerate orbitals will start only after each of these orbitals is occupied by $1e^{\Theta}$ of same spin.

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Space for Rough Work

43. Among the following transition metal ions, the one where all the metal ions have $3d^2$ electronic configuration is [Z : Ti = 22, V = 23, Cr = 24, Mn = 25]
(A) Ti^{3+}, V^{2+}, Cr^{3+}, Mn^{4+} (B) Ti^{4+}, V^{4+}, Cr^{+6}, Mn^{7+} (C) Ti^{4+}, V^{3+}, Cr^{2+}, Mn^{3+} (D) Ti^{2+}, V^{3+}, Cr^{4+}, Mn^{5+}
(E) Ti^{3+}, V^{3+}, Cr^{3+}, Mn^{3+}

44. In a hypothetical situation, if a body absorbs a photon of wavelength λ and emits photons of wave length $2\lambda_0$, $4\lambda_0$, $8\lambda_0$, $16\lambda_0$, ……. etc so that the body again acquires the original energy. Given $\lambda = x\lambda_0$ then find x.
(A) 1 (B) 2 (C) 3 (D) 4 (E) 1/2

45. Calculate the number of total orbitals present in the shell which would be the first to have ' f ' sub shell :
(A) 9 (B) 16 (C) 25 (D) 36 (E) 4

46. Electron in He^+ is in a state with total 1 node with zero orbital angular momentum. Photons of wavelength $\frac{9}{5R}$ cause the electron to jump into a higher state. The higher state in which electron jumps finally is associated with five possible orientations in space. What is total number of radial nodes in higher state ? (R = Rydberg's constant)
(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

47. If uncertainty in momentum is twice the uncertainty in position of an electron, then uncertainty in velocity of electron is $\frac{x}{m}\sqrt{\hbar}$. Than value of x is [$\hbar = \frac{h}{2\pi}$]
(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

48. He^+ ions are in excited state from where maximum 6 types of photons can be emitted. If E is the separation energy of electron in this excited state of He^+ ion, then how many times of energy E is required to ionize hydrogen atom ?
(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

49. The degeneracy of 4^{th} excited state of ${}_5B^{3+}$ ion is ______.
(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

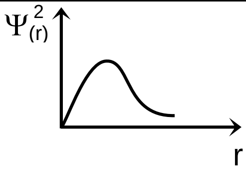
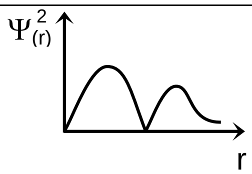
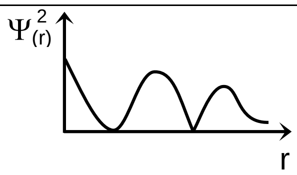
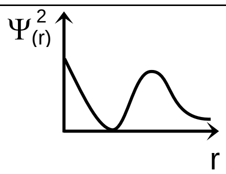
50. The maximum number of electrons in ${}_{13}A\ell$ for which $\frac{\ell x m}{n} = 0$ is ___________.
(A) 7 (B) 8 (C) 9 (D) 10 (E) 6

Space for Rough Work

SECTION – 3 : (Maximum Marks : 12)

- This section contains **TWO (02)** paragraphs
- Based on each paragraph, there will be **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four option is correct
- Marking scheme :
  - Full Marks : **+3** If ONLY the correct option is chosen.
  - Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).
  - Negative Marks : **–1** In all other cases

Paragraph for Question Nos. 51 to 52

Column-I		Column-II		Column-III	
(I)	$\Psi_{(r)} = \frac{1}{2\sqrt{2}}\left(\frac{z}{a_0}\right)^{3/2}(2-\sigma)e^{-\frac{\sigma}{2}}$	(i)		(P)	Total number of nodes = 2
(II)	$\Psi_{(r)} = \frac{1}{9\sqrt{3}}\left(\frac{z}{a_0}\right)^{3/2}(6-6\sigma+\sigma^2)e^{-\sigma/2}$	(ii)		(Q)	Total number of nodes = 1
(III)	$\Psi_{(r)} = \frac{1}{9\sqrt{6}}\left(\frac{z}{a_0}\right)^{3/2}(4-\sigma)\sigma e^{-\sigma/2}$	(iii)		(R)	Number of angular nodes = 2
(IV)	$\Psi_{(r)} = \frac{1}{9\sqrt{30}}\left(\frac{z}{a_0}\right)^{3/2}\sigma^2 e^{-\sigma/2}$	(iv)		(S)	Number of angular nodes = 1

Space for Rough Work

51. Which of the following is the correct combination :

(A) (I) (iv) (Q) (B) (II) (ii) (P) (C) (III) (iv) (S) (D) (I) (ii) (R)

52. Which of the following is the incorrect combination :

(A) (III) (ii) (P) (B) (II) (iii) (P) (C) (III) (ii) (S) (D) (IV) (ii) (R)

Paragraph for Question Nos. 53 to 54

The hydrogen like species He^+ is in a spherically symmetric state S_1 with two radial nodes. Upon absorbing light, the ion undergoes transition to a state S_2 which has two radial nodes and has energy equal to $\frac{1}{4}$ times the ground state energy of H-atom.

Now answer the following questions :

53. The value of orbital angular momentum for an electron in S_2 state is :

(A) 0 (B) $\frac{\sqrt{6}\,h}{2\pi}$

(C) $\frac{\sqrt{2}\,h}{2\pi}$ (D) Cannot be determined due to data insufficiency

54. How much energy must have been absorbed for electronic transition from S_1 state to S_2 state :

(A) 2.64 eV (B) 10.2 eV

(C) 3.864 eV (D) Cannot be determined due to data insufficiency

Space for Rough Work